

Integer Arithmetic

Easy

- NOP $\rightarrow 0$
- ADD
- SUB
- MOV
- AND
- OR
- XOR
- NOT
- SHL
- SHR
- SHA (shift arithmetic)
- MIN (signed)
- MAX (signed)
- ABS (Absolute value)
- ???
- MUL
- MAD/MAC
- NEG (negate)
- DIV
- REM (remainder)

27 immediately needed

18 non-trivial instructions

34 overall required

rd, r1, r2, r3, imm

Comparison:

- SLT
- SEQ
- SLE

Special Function Units

- RCP ($1/x$) 23
- RSQRT ($1/\sqrt{x}$) 24
- SQRT 25

- everything else is too complicated/niche utility on

Memory Operations

???

Control Flow

- BEQ 26
- BNE 27

Not relevant yet but needed

- SSY 28
- JOIN 29
- BAR 30
- TID 31
- CALL 32
- RET 33
- Exit 34